

Introduction

This document explains the StoryStacker™ asset management pipeline, runtime size optimization strategies, and

1. Pipeline Overview

The asset pipeline consists of:

1. Incoming Folder - Artists drop new files here.
2. `asset_drop.sh` - A script that processes incoming assets:
 - Normalizes file names.
 - Routes assets to the correct source folders.
 - Compresses and converts assets into runtime-ready formats.
 - Generates LODs for models and optimized texture formats.
3. Assets Folder - Contains all source, high-resolution assets.
4. Runtime Folder - Contains optimized, compressed game-ready assets.
5. Reports Folder - Contains size reports and budget checks.
6. Paks Folder - Contains theme-based downloadable content packs.

2. Strategies to Keep Build Size Small

- Use texture compression formats (KTX2, ASTC, BCn).
- Use shared texture atlases and trim sheets.
- Use mesh compression (Draco/meshopt) and Level of Detail (LOD) models.
- Deduplicate textures and materials by hashing.
- Stream large audio and video files.
- Use theme-based downloadable content packs.

3. GitHub Actions Workflow

The provided GitHub Actions workflow will:

- Automatically run `asset_drop.sh` when you push or create a PR.
- Generate runtime size reports.
- Enforce size budgets (fails if total > 7 GB or per-asset > 600 KB).
- Publish a size dashboard to GitHub Pages.
- Build theme-based PAK files.

4. Runtime Size Dashboard

The dashboard is a static HTML file that reads the size report and shows:

- Total install size.
- Number of assets.
- Per-asset sizes (sortable and filterable).

It allows you to quickly identify large assets and manage game build size.

5. GitHub Setup Instructions

Follow these steps to set up GitHub for StoryStacker™:

1. Create a new GitHub repository or use your existing one.
2. Create the following folders at the root of the repo:
 - incoming
 - Assets
 - Runtime
 - Reports
 - Paks
3. Add the provided `asset_drop.sh` script to the root of your repo.
4. Add the `.github/workflows/runtime-bake.yml` file for GitHub Actions.

6. Using the Pipeline to Develop StoryStacker™

- Artists place files in /incoming with naming convention theme_assetid_optional.ext.
- Run asset_drop.sh locally or via GitHub Actions.
- Compressed runtime assets are placed in /Runtime for the game.
- Check /Reports for size compliance and budget alerts.
- Use /Paks for theme-based downloadable content.
- Review the dashboard regularly to ensure size targets are met.